Regression Assignment

1) Identify your Problem Statement:

Machine Learning → Supervised Learning → Regression

2) Basic Information About the Dataset:

Total Number of Rows: 1338

Total Number of Columns: 6

Columns:

age - Age of the individual

sex - Gender of the individual

bmi - Body Mass Index

children - Number of children/dependents

smoker - Smoking status

charges - Medical insurance charges

3) Pre-processing Method:

One-Hot Encoding was used using `pd.get_dummies()`.

The categorical columns:

sex → sex_female, sex_male

smoker → smoker_no, smoker_yes

After preprocessing, the dataset contains 8 columns (7 input features + 1 target column).

4) Develop a Good Model:

Algorithms trained:

Linear Regression

Support Vector Regressor (SVR)

Decision Tree Regressor

Random Forest Regressor

5) Research Values (R2 Score):

Algorithm	R2 Score
Linear Regression	0.75
Support Vector Regressor (SVR)	0.65
Decision Tree Regressor	0.71
Random Forest Regressor	0.86

6) Final Model & Justification:

Selected Model: Random Forest Regressor

Justification: Achieved the highest R2 score (~0.86) and accurately captures non-linear interactions (like bmi and smoker).